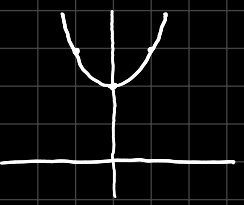
1. $f(x) = x^2 + 2$

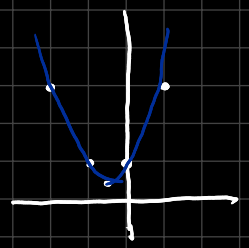


x-intercepts: None
y-intercepts: $(0,2)$

$f(x) = x^2 + 2 = 1(x+0)^2 + 2$

D: $(-\infty, \infty)$ Increasing: $[0, \infty)$ Vertex: $(0,2)$ Absolute Minimum
R: $[2, \infty)$ Decreasing: $(-\infty, 0]$ Symmetry: y-axis

7. $f(x) = x^2 + x + 1$



x-intercept: None
y-intercept: $(0,1)$

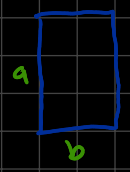
$f(x) = x^2 + x + 1 = 1(x + 1/2)^2 + 0$

D: $(-\infty, \infty)$ Increasing: $[-1/2, \infty)$ Vertex: $(-1/2, 3/4)$ Absolute Minimum
R: $[3/4, \infty)$ Decreasing: $(-\infty, -1/2)$ Symmetry: $x = -1/2$

15. $C(x) = 0.1x + 25$, $p(x) = 10 - 0.01x$, $P(x) = R(x) - C(x) = xp(x) - C(x) = x(10 - 0.01x) - (0.1x + 25) = 10x - 0.01x^2 - 0.1x - 25$
$= -0.01x^2 + 9.9x - 25$

Vertex: $\left(\frac{99}{0.01}, P\left(\frac{99}{0.01}\right)\right) = (495, 2425.25)$, max profit with 495 cookies

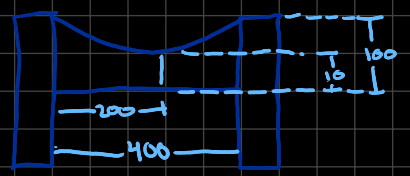
21.



$\rightarrow A = ab$, $2a + 2b = 14 \rightarrow a = \frac{14 - 2b}{2} = 7 - b$, $A = (7-b)b = 7b - b^2$, $A' = 7 - 2b$, $7 - 2b = 0$, $b = 7/2$

$A(7/2) = 7(7/2) - (7/2)^2 = 49/4 = 12.25$

26.



$f(x) = ax^2 + bx + c$
$f(200) = 10$ $c = 100$
$f(0) = 100$ $(200)^2 a + 200b + c = 10$ $\rightarrow$ $a = \frac{9}{4000}$
$f(400) = 100$ $(400)^2 a + 400b + c = 100$ $b = -9/10$
$c = 100$

$f(x)\ \frac{9}{4000}x^2 - \frac{9}{10}x + 100$, $f(50) = 60.625$

31. $x^2 - 10y^2 = 0$, $x^2 = 10y^2$, $x = \pm\sqrt{10y^2}$

35. $y^2 - 4y = x^2 - 4$, $y^2 - 4y - (x^2 - 4) = 0$ $y = \frac{+4 \pm \sqrt{16 + 4(x^2 - 4)}}{2} = \frac{4 \pm \sqrt{4x^2}}{2} = 2 \pm x$